CHAPTER

13

Y-Chromosome DNA Testing

LINEAGE MARKERS

Autosomal DNA markers, such as the short tandem repeat (STR) loci discussed in Chapter 5 or the single-nucleotide polymorphisms (SNPs) covered in Chapter 12, are shuffled with each generation because half of an individual's genetic information comes from his/her father and half from his/her mother. However, the Y-chromosome (ChrY) and mitochondrial DNA (mtDNA) markers that will be discussed in this chapter and Chapter 14 represent "lineage markers." They are passed down from generation to generation without changing (except for mutational events). Maternal lineages can be traced with mitochondrial DNA sequence information while paternal lineages can be followed with Y-chromosome markers (Figure 13.1).

With lineage markers, the genetic information from each marker is referred to as a haplotype rather than a genotype because there is usually only a single allele per individual. Because Y-chromosome markers are linked on the same chromosome and are not shuffled with each generation, the statistical calculations for a random match probability cannot involve the product rule. Therefore, haplotypes obtained from lineage markers can never be as effective in differentiating between two individuals as genotypes from autosomal markers that are unlinked and segregate separately from generation to generation. However, Y-chromosome (this chapter), mitochondrial DNA (Chapter 14), and X-chromosome (Chapter 15) markers can play an important role in forensic investigations as well as other human identification applications.

APPLICATIONS OF ChrY TESTING

Y-chromosome DNA testing is important for a number of different applications of human genetics including forensic evidence examination, paternity testing, historical investigations, studying human migration patterns throughout history, and genealogical research. In terms of forensic applications, there are both advantages and limitations to Y-chromosome testing (Table 13.1).

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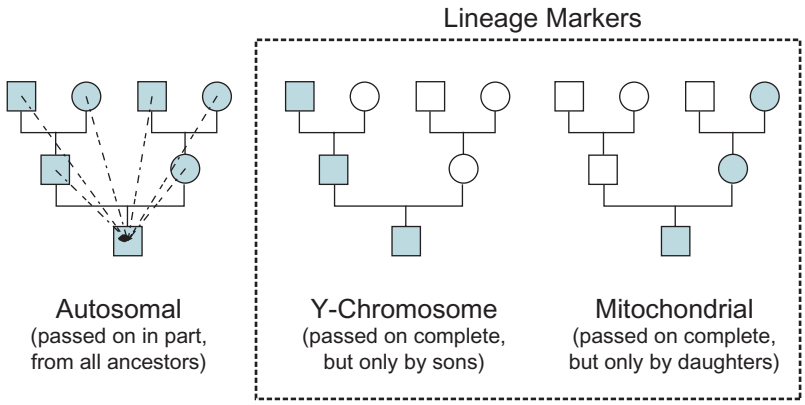


FIGURE 13.1 Illustration of inheritance patterns from recombining autosomal genetic markers and the lineage markers from the Y-chromosome and mitochondrial DNA.

TABLE 13.1 Advantages and Limitations of Y-Chromosome DNA Testing.

Advantages	Limitations
1. Male-specific amplification can enable examination of a male perpetrator's profile even in mixtures with high levels of female DNA in sexual assault cases. 2. Additional mixtures may possibly be analyzed (e.g., fingernail scrapings, saliva on skin, etc.) 3. Paternal transmission from a father to all of his sons extends possible reference sample providers and enables tracing family lineages.	1. Since paternal relatives are identical, Y-STR typing cannot be used to distinguish among brothers or even distant paternal relatives. 2. Without recombination between loci, the product rule cannot be used and thus the discrimination power of Y-STRs is limited by the size of the population database used. 3. Duplications and deletions can complicate the analysis.

The primary value of the Y-chromosome in forensic DNA testing is that it is found only in males. The SRY (sex-determining region of the Y) gene determines maleness. Since a vast majority of crimes where DNA evidence is helpful, particularly sexual assaults, involve males as the perpetrators, DNA tests designed to only examine the male portion can be valuable.

With ChrY tests, interpretable results can be obtained in some cases where autosomal tests are limited by the evidence, such as high levels of female DNA in the presence of minor amounts of male DNA (Figure 13.2). These situations include sexual assault evidence from azospermic or vasectomized males and blood–blood or saliva–blood mixtures where the absence of sperm prevents a successful differential extraction for isolation of male DNA (Prinz & Sansone 2001). In addition, the number of individuals involved in a "gang rape" may be easier to decipher with Y-chromosome results than with highly complicated autosomal STR mixtures. Using ChrY-specific PCR primers can improve the chances of detecting low levels of the perpetrator's DNA in a high background of a female victim's DNA (Hall & Ballantyne 2003). Y-chromosome tests have also been used to verify amelogenin Y-deficient males (Thangaraj et al. 2002).

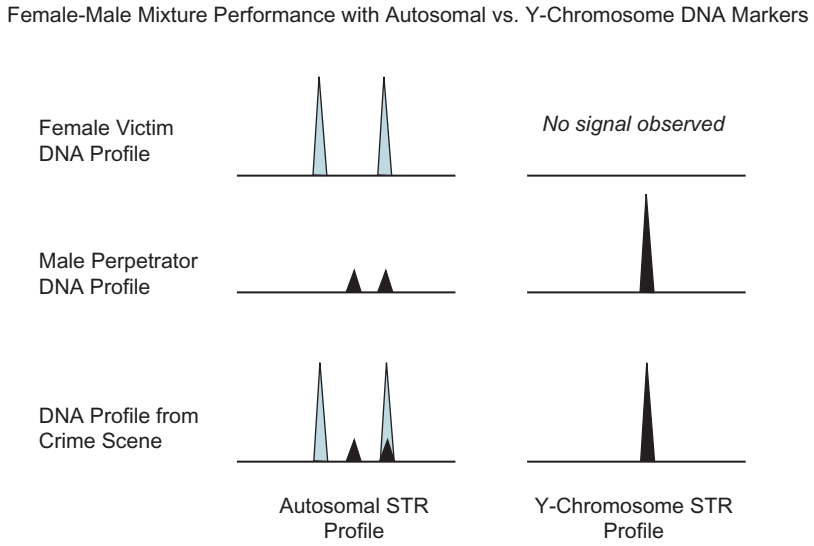


FIGURE 13.2 Schematic illustrating the types of autosomal or Y-STR profiles that might be observed with sexual assault evidence where mixtures of high amounts of female DNA may mask the STR profile of the perpetrator. In some cases, Y-STR testing permits isolation of the male component without having to perform a differential lysis.

The same feature of the Y-chromosome that gives it an advantage in forensic testing, namely maleness, is also its biggest limitation. A majority of the Y-chromosome is transferred directly from father to son without recombination to shuffle its genes and provide greater genetic variety to future generations. Random mutations are the only mechanisms for variation over time between paternally related males. Thus, while exclusions in Y-chromosome DNA testing results can aid forensic investigations, a match between a suspect and evidence only means that the individual in question could have contributed the forensic stain—as could a brother, father, son, uncle, paternal cousin, or even a distant cousin from his paternal lineage (Figure 13.3)! Needless to say, inclusions with Y-chromosome testing are not as meaningful as autosomal STR matches from a random match probability point-of-view (de Knjiff 2003).

On the other hand, the presence of relatives having the same ChrY (Figure 13.3) expands the number of possible reference samples in missing persons investigations and mass disaster victim identification efforts. ChrY testing also aids familial searching (Dettlaff-Kakol & Pawlowski 2002, Sims et al. 2008). Deficient paternity tests where the father is dead or unavailable for testing are benefited if ChrY markers are used (Santos et al. 1993). However, an autosomal DNA test is always preferred when possible since it provides a higher power of discrimination.

The Y-chromosome has also become a popular tool for tracing historical human migration patterns through male lineages (Jobling & Tyler-Smith 1995, 2003). Anthropological, historical, and genealogical questions can be answered through ChrY results. For example, ChrY results in 1998 linked modern-day descendants of Thomas Jefferson and Eston Hemings leading to the controversial conclusion that Jefferson fathered the slave (Foster et al. 1998).

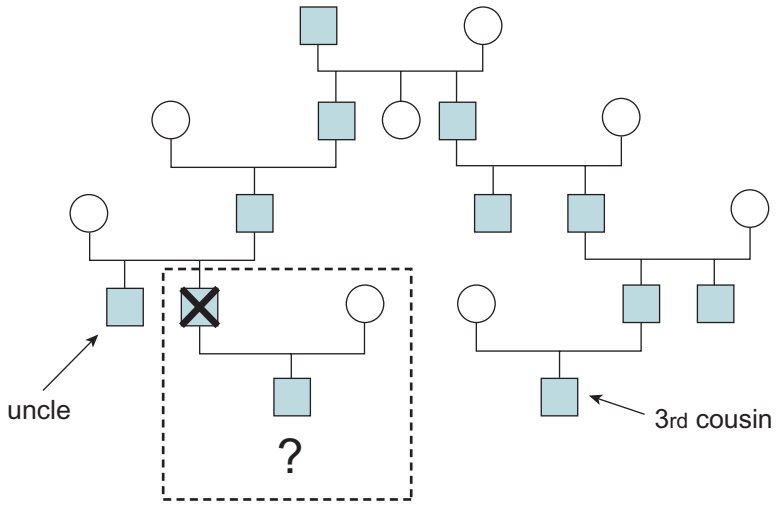


FIGURE 13.3 An example pedigree showing patrilineal inheritance where all shaded males have the same Y-chromosome barring any mutations. To help identify the person in question, any of the other males with the same patrilineage could provide a reference sample to assist in a missing persons investigation, mass disaster victim identification, or deficient paternity test (boxed region) where the father is deceased or not available for testing.

ChrY STRUCTURE

A detailed analysis of the "finished" reference Y-chromosome sequence was described in the June 19, 2003 issue of *Nature* by researchers from the Whitehead Institute and Washington University. Although it is stated as being a "finished" sequence, Skaletsky et al. (2003) report on only 23 Mb of the roughly 50 Mb present in a typical human ChrY.

The unreported and as yet unsequenced ≈30 Mb portion is a heterochromatin region located on the long arm of the Y-chromosome (Figure 13.4) that is not transcribed and is composed of highly repetitive sequences, which are impossible to sequence reliably with current technology. At 50 Mb, the Y-chromosome is the third smallest human chromosome—only slightly larger than chromosome 21 (47 Mb) and chromosome 22 (49 Mb).

The tips of the Y-chromosome, which are called the *pseudoautosomal regions* (PAR), recombine with their sister sex X-chromosome homologous regions. PAR1 located at the tip of the short arm (Yp) of the Y-chromosome is approximately 2.5 Mb in length while PAR2 at the tip of the long arm (Yq) is less than 1 Mb in size (Graves et al. 1998). The remainder of the Y-chromosome (≈95%) is known as the *non-recombining portion of the Y-chromosome*, or NRY (Figure 13.4a). The NRY remains the same from father to son unless a mutation occurs.

Some authors term the NRY the male-specific region (MSY) because of evidence of frequent gene conversion or intrachromosomal recombination (Skaletsky et al. 2003). A total of 156 known transcription units including 78 protein-coding genes are present on MSY.

Many sequences in the ChrY are highly duplicated either with themselves or with the X-chromosome. Three classes of sequences have been characterized in the Y-chromosome: X-transposed, X-degenerate, and ampliconic (Skaletsky et al. 2003). Two blocks on the short

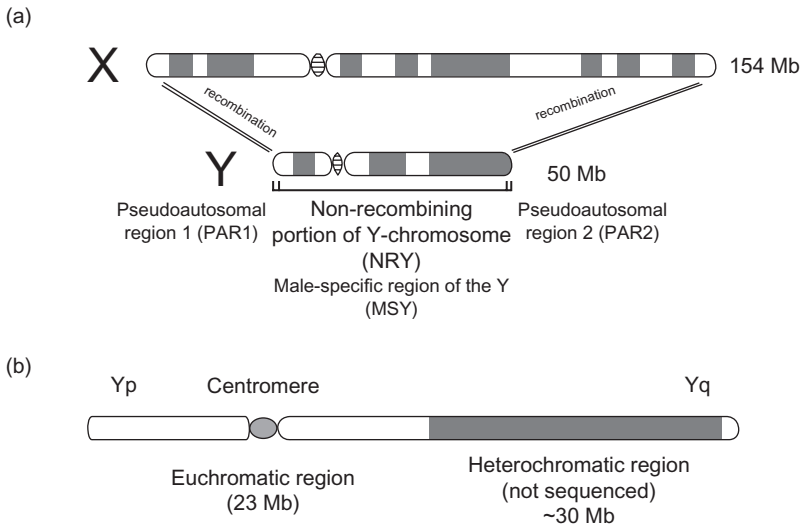


FIGURE 13.4 (a) Schematic of X and Y sex chromosomes. The two tips of the Y-chromosome known as pseudoautosomal region 1 (PAR1) and 2 (PAR2) recombine with the tips of the X-chromosome. The remaining 95% of the Y-chromosome is referred to as the non-recombining portion of the Y-chromosome (NRY) or male-specific region of the Y (MSY). (b) The Y-chromosome is composed of both euchromatic and heterochromatic regions of which only the 23 Mb of euchromatin has been sequenced.

arm of ChrY with a combined length of 3.4 Mb make up the X-transposed sequences. These sequences are 99% identical to sequences found in Xq21, contain two coding genes, and do not participate in X–Y crossing over during male meiosis.

X-degenerate segments of MSY occur in eight blocks on both the short arm and the long arm of the Y-chromosome with an aggregate length of 8.6 Mb. These X-degenerate segments possess up to 96% nucleotide sequence identity to their X-linked homologues. These X-homologous regions can make it challenging to design Y-chromosome assays that generate male-specific DNA results. If portions of an X-homologous region of the Y-chromosome are examined inadvertently, then female DNA, which possesses two X-chromosomes, will be detected. Thus, when testing Y-chromosome-specific assays it is important to examine them in the presence of female DNA (high levels) to verify that there is little-to-no cross talk with X-homologous regions of the Y-chromosome (Butler et al. 2002, Hall and Ballantyne 2003).

The ampliconic segments are composed of seven large blocks scattered across both the short arm and the long arm and covering about 10.2 Mb of the Y-chromosome (Skaletsky et al. 2003). Some 60% of these ampliconic sequences have intrachromosomal identities of 99.9% or greater. In other words, it is very difficult to tell these sequences apart from one another. Another interesting feature of these ampliconic segments is that many of them are palindromes—that is, the almost exact duplicate sequences are inverted with respect to each other's sequence essentially as mirror images. Eight large palindromes collectively comprise 5.7 Mb of Yq with at least six of these palindromes containing testis genes. Genetic markers within these palindromic regions will exist as multi-copy PCR products from single primer

sets. For example, the DAZ (deleted in azospermic) gene occurs in four copies at ≈24 Mb along the reference sequence (Saxena et al. 1996, Skaletsky et al. 2003).

Different Classes of ChrY Genetic Markers

Two broad categories of DNA markers have been used to examine Y-chromosome diversity: bi-allelic loci, which exhibit two possible alleles, and multi-allelic loci. Results from typing the lower resolution bi-allelic markers are classified into haplogroups while multi-allelic results are characterized as haplotypes (de Knijff 2000).

Bi-allelic markers include single nucleotide polymorphisms (Y-SNPs) and an *Alu* element insertion. The Y-Alu polymorphism (YAP) was the first discovered Y-chromosome bi-allelic marker (Hammer 1994). Bi-allelic markers are sometimes referred to as unique event polymorphisms (UEPs) because of their low mutation rates (≈10^{-8} to 10^{-9} per generation). More than 600 bi-allelic Y-chromosome markers have now been characterized into defined haplogroups (Y Chromosome Consortium 2002, Butler 2003, Karafet et al. 2008).

Y-chromosome multi-allelic markers include two minisatellites and several hundred short tandem repeat (Y-STR) markers (Kayser et al. 2004). These multi-allelic loci can be used to differentiate Y-chromosome haplotypes with fairly high resolution due to their higher mutation rates. Minisatellite loci have mutation rates as high as 6% to 11% per generation (Jobling et al. 1999) while the average mutation rate for Y-STRs is ≈0.2% per generation (Kayer et al. 2000, Dupuy et al. 2004).

Y-STR MARKERS

Y-chromosome DNA analysis can be performed with either Y-STRs or Y-SNPs. Since Y-STRs change more rapidly (mutation rate ≈1 in 10^3) compared to Y-SNPs (mutation rate ≈1 in 10^9), Y-STR results exhibit more variability and thus have greater use in forensic applications. Typically Y-STRs are described as defining haplotypes while Y-SNPs define haplogroups. As will be discussed at the end of the chapter, Y-SNPs can be useful in DNA ancestry studies.

Single Copy vs. Multi-Copy Markers

Due to the duplicated, palindromic regions of the Y-chromosome mentioned above, some Y-STR loci occur more than once and, when amplified with a locus-specific set of primers, produce more than one PCR product. This duplication can lead to some confusion in terms of counting the number of loci present in a haplotype. A single set of primers can produce two amplicons, which may be thought of as "two loci" for a Y-chromosome haplotype.

For example, the Y-STR locus DYS385 is present in two regions along the long arm of the Y-chromosome. These duplicated regions are located about 40,000 bp apart and can generate two different alleles when amplified with a single set of primers. The two alleles are typically labeled "a" and "b" with the "a" designation going to the smaller-sized allele. It is also possible to have both "a" and "b" alleles be the same size in which case only a single peak would appear in an electropherogram (Figure 13.5a). Due to the presence of two

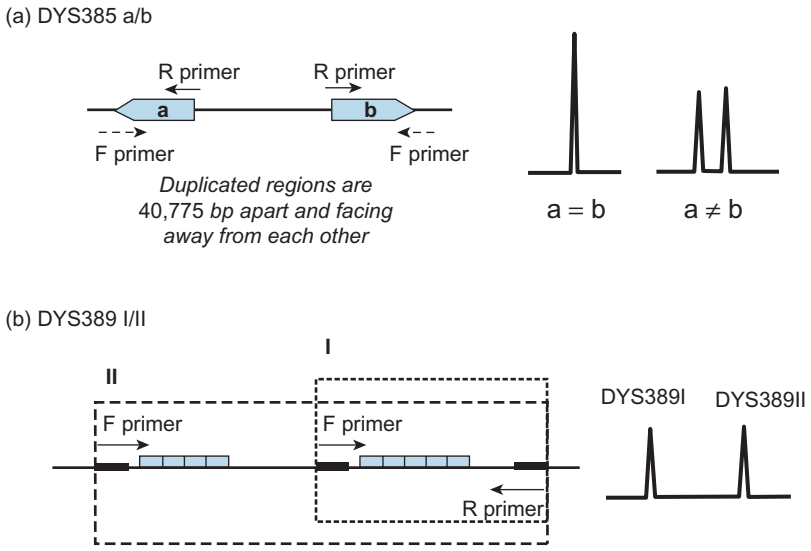


FIGURE 13.5 Schematic illustration of how multiple PCR primer binding sites give rise to multi-copy PCR products for (a) DYS385a/b and (b) DYS389I/II. Arrows represent either forward "F" or reverse "R" primers. In the case of DYS385a/b, the entire region around the STR repeat is duplicated and spaced about 40,775 bp apart on the long arm of the Y-chromosome. Thus, amplification with a single set of primers gives rise to one peak if the "a" repeat region is equal in size to the "b" repeat region or separate peaks if "a" and "b" differ in length. DYS389 possesses two primary repeat regions that are flanked on one side by a similar sequence. Widely used forward primers bind adjacent to both repeats generating amplicons that differ in size by ≈120 bp. Note that DYS389II is inclusive of the DYS389I repeat region and therefore some analyses subtract DYS389II–DYS389I repeats.

alleles, this duplicated locus is usually referred to as DYS385 a/b. It has been demonstrated that with a nested PCR approach, the "a" and "b" alleles for DYS385 can be amplified separately (Kittler et al. 2003, Seo et al. 2003). Other multi-copy Y-STRs besides DYS385 a/b that have been used in human identity testing include YCAII a/b and DYS464 a/b/c/d (Redd et al. 2002, Butler 2003, Schoske et al. 2004).

Two PCR products can also be generated at the DYS389 locus with a single set of primers. However in this case the DYS389I PCR product is a subset of the DYS389II amplicon because the forward PCR primer binds to the flanking region of two different repeat regions that are approximately 120 bp apart (Figure 13.5b). Some analyses with DYS389I/II treat the larger PCR product as DYS389II–DYS389I to get a handle on the variation occurring in the two regions independent of each another (e.g., Redd et al. 2002).

Minimal Haplotype Loci

The number of Y-STR loci available for use in human identity testing has increased dramatically since the turn of the century and the availability of the human genome sequence. In the 1990s only a handful of Y-STR markers were characterized and available for use. At the beginning of 2002, only about 30 Y-STRs were available for researchers (Butler 2003).

TABLE 13.2 Characteristics of Commonly Used Y-Chromosome STR Loci. Adapted from Butler (2006) and Decker et al. (2007). Mutation Rates Are from as Many as 15000 Meioses Described in a YHRD Summary of 23 Publications as of January 2011 (See http://www.yhrd.org/Research/Loci/).

STR Marker	Position (Mb)	Repeat Motif	Allele Range	Mutation Rate
DYS393	3.19	AGAT	8–17	0.10%
DYS456	4.33	AGAT	13–18	0.42%
DYS458	7.93	GAAA	14–20	0.64%
DYS19	10.13	TAGA	10–19	0.23%
DYS391	12.61	TCTA	6–14	0.26%
DYS635	12.89	TSTA	17–27	0.35%
DYS437	12.98	TCTR	13–17	0.12%
DYS439	13.03	AGAT	8–15	0.52%
DYS389 I/II	13.12	TCTR	9–17/24–34	0.25%/0.36%
DYS438	13.38	TTTTC	6–14	0.03%
DYS390	15.78	TCTR	17–28	0.21%
GATA-H4	17.25	TAGA	8–13	0.24%
DYS385 a/b	19.26	GAAA	7–28	0.21%
DYS392	21.04	TAT	6–20	0.04%
DYS448	22.78	AGAGAT	17–24	0.16%

Since that time more than 400 new Y-STRs (Kayser et al. 2004, Leat et al. 2007) have been discovered and deposited in the Genome Database (GDB). These Y-STRs have been cataloged and mapped to their ChrY positions (Hanson & Ballantyne 2006). Unfortunately, operations with the GDB were halted in 2008 and thus this repository of DNA marker information is no longer available online.

Yet even with a limited number of loci available at the time, a core set was selected in 1997 that continue to serve as "minimal haplotype" loci (Kayser et al. 1997, Pascali et al. 1998). The minimal haplotype is defined by the single copy Y-STR loci DYS19, DYS389I, DYS389II, DYS390, DYS391, DYS392, DYS393, and the highly polymorphic multi-copy locus DYS385 a/b (Schneider et al. 1998). By means of a multicenter study, more than 4000 male DNA samples from 48 different subpopulation groups were studied with the single copy loci in the minimal haplotype set (de Knijff et al. 1997). This work formed the basis for what is now the online Y-STR Haplotype Reference Database (http://www.yhrd.org) that will be described in more detail below.

In January 2003, the U.S. Scientific Working Group on DNA Analysis Methods (SWGDAM) recommended use of the minimal haplotype loci plus two additional single-copy Y-STRs: DYS438 and DYS439 (Ayub et al. 2000). Information regarding these core loci and other loci present in commercial Y-STR kits may be found in Table 13.2. Their position along ChrY is illustrated in Figure 13.6. Although other Y-STRs may be added to databases

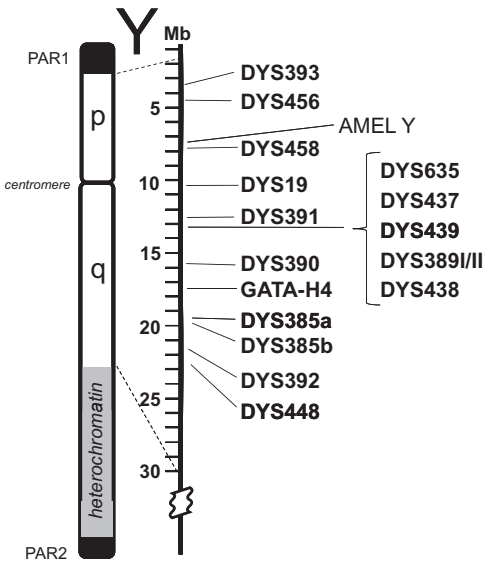


FIGURE 13.6 Relative positions of 17 Y-STR loci commonly used in ChrY testing. DYS458 is closest to amelogenin (AMEL Y) and thus may be lost with AMEL Y null alleles due to deletion of that portion of ChrY.

as their value is demonstrated and they become part of commercially available kits, the original minimal haplotype loci and SWGDAM recommended Y-STRs are likely to dominate human identity applications in the coming years.

Y-STR Kits

As noted in Chapter 5, forensic scientists rely heavily on commercially available kits to perform DNA testing. Thus, many laboratories especially in the U.S. were reluctant to move into Y-STR typing until Y-STR kits were offered. Table 13.2 lists the loci present in the two most widely used Y-STR kits (Figure 13.7): PowerPlex Y (Promega Corporation) and Yfiler (Applied Biosystems). Note that all of the European and U.S. core Y-STR loci are included in both kits. PowerPlex Y contains one additional locus (DYS437) and Yfiler has six additional loci (DYS437, DYS448, DYS456, DYS458, DYS635, and GATA-H4). Yfiler kit allelic ladders are shown in Figure 13.8.

Until 2005, ReliaGene Technologies (formerly of New Orleans, LA) sold the Y-PLEX 12 kit, which amplified the SWGDAM recommended loci plus the amelogenin marker (see Chapter 5). Reliagene had also supplied Y-PLEX 6 and Y-PLEX 5 kits (Sinha et al. 2004), which were precursors to the Y-PLEX 12 kit. Inclusion of amelogenin enables confirmation that the PCR reaction has not failed on female DNA samples since a single X amplicon will result. In addition, mixture levels of male and female DNA can be confirmed in many situations with the amelogenin X and Y peak height ratios. While the amelogenin primers provide a measure of quality control on PCR amplifications, they have the disadvantage of possibly tying up and consuming PCR reagents when high levels of female DNA are present in a mixture.

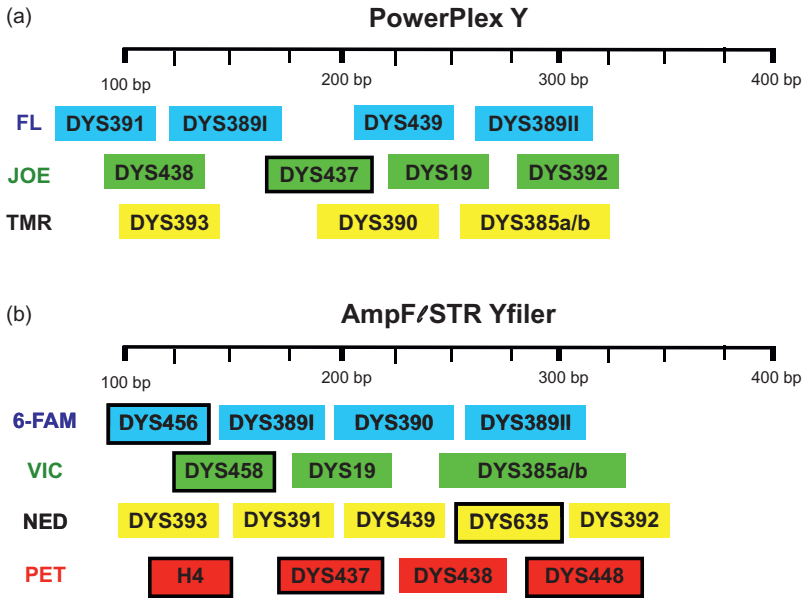


FIGURE 13.7 Comparison of relative size ranges and fluorescent dye labels for Y-STR loci present in (a) PowerPlex Y from Promega Corporation and (b) Yfiler from Applied Biosystems. The highlighted locus boxes are those additional loci included in the multiplex beyond the SWGDAM-recommended set.

Y-STR Allele Nomenclature

The DNA Commission of the International Society of Forensic Genetics (ISFG) has made a series of recommendations on the use of Y-STR markers (Gill et al. 2001, Gusmão et al. 2006). Their recommendations address allele nomenclature, use of allelic ladders, population genetics, and reporting methods.

The ISFG recommendations for Y-STR allelic ladders include the following: (a) the alleles should span the distance of known allelic variants for a particular locus, (b) the rungs of the ladder should be one repeat unit apart wherever possible, (c) the alleles present in the ladder should be sequenced, and (d) the ladders should be widely available to enable reliable interlaboratory comparisons. The existence of commercially available Y-STR kits has now facilitated the widespread use of consistent allelic ladders (e.g., Figure 13.8).

Prior to commercially available Y-STR kits and consistent allelic ladders, various researchers in the field took different approaches to naming alleles. For some loci there were instances of multiple published designations for the same allele. An example of this phenomenon that illustrates the importance of standardization is DYS439, which has been designated three different ways in the literature (Figure 13.9).

In an effort to provide a unified nomenclature for STR loci, a comparative analysis of the repeat and sequence structure of Y-chromosome markers in humans and chimpanzees has been proposed and 11 human Y-STRs have been studied (Gusmão et al. 2002). Since the chimpanzees examined in their study did not vary in the other regions outside of the variable core GATA repeat for DYS439, Gusmão and co-workers (2002) proposed a $[GATA]_n$

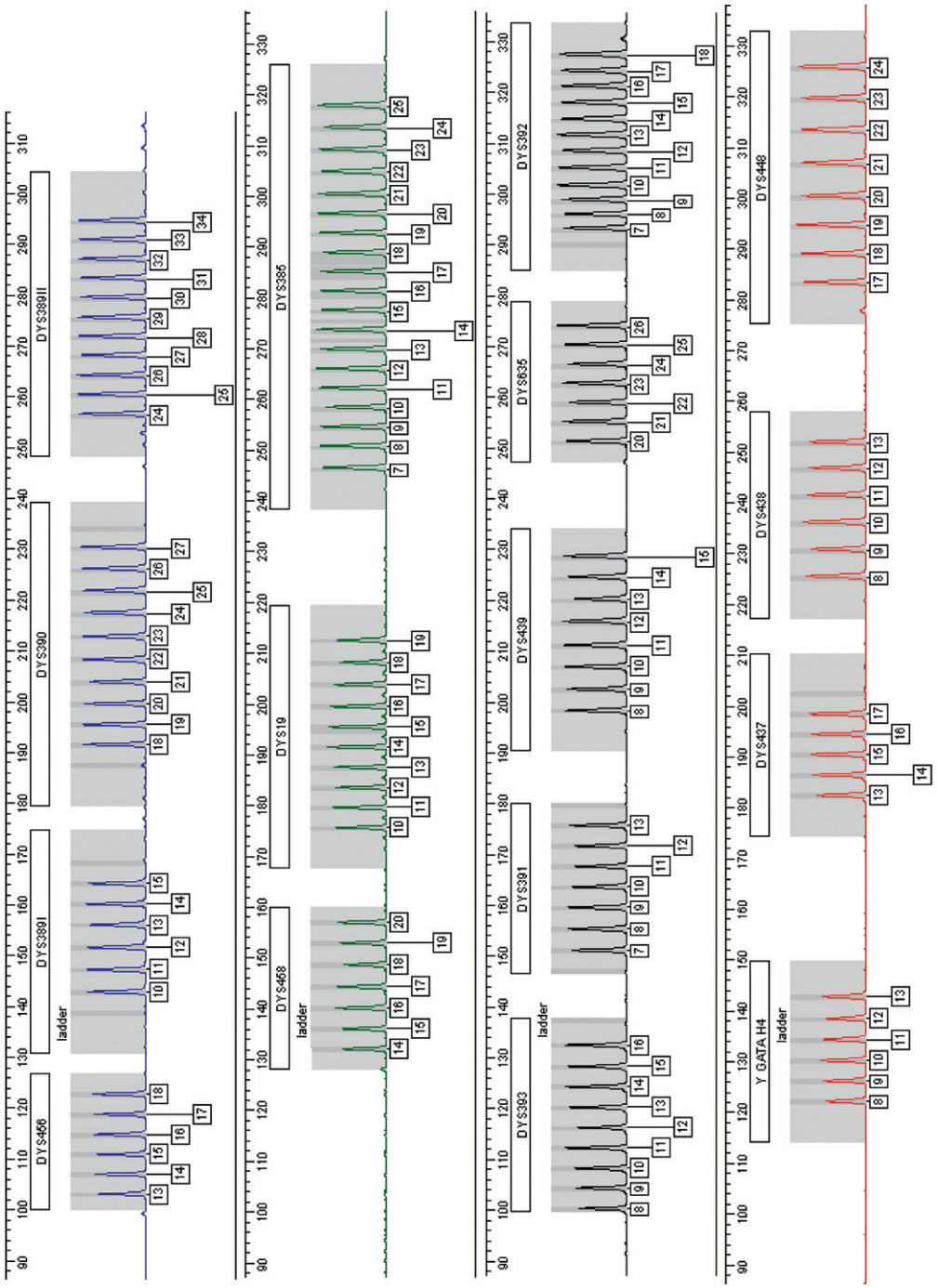


FIGURE 13.8 Yfiler kit allelic ladders.

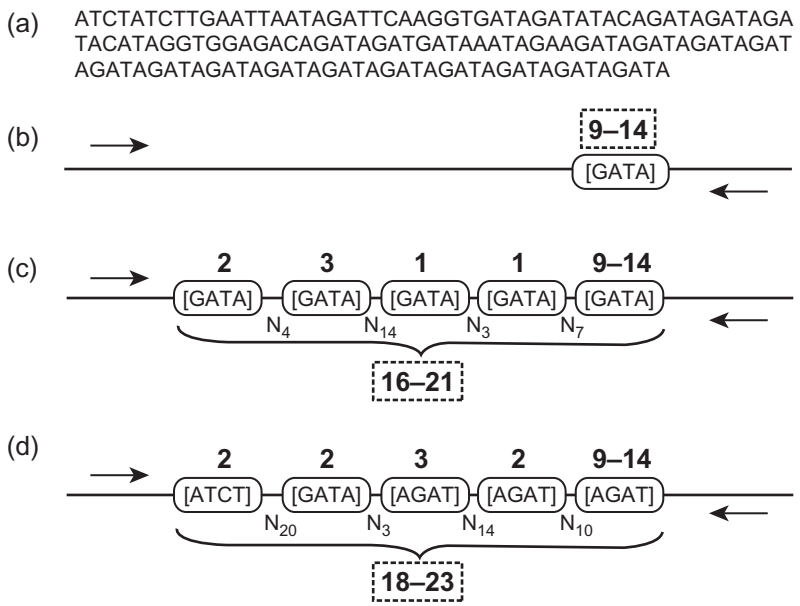


FIGURE 13.9 Various published allele nomenclatures for DYS439. (a) Sequence from DYS439 spanning the repeat regions used in the various nomenclatures; (b) Schematic of allele designation by Ayub et al. (2000)—repeat range is 9–14; (c) Schematic of allele designation by Grignani et al. (2000)—repeat range is 16-21; (d) Schematic of allele designation by Gonzalez-Neira et al. (2001)—repeat range is 18–23. The most widely accepted designation and what is used in commercial Y-STR kits is (b)—that of Ayub et al. (2000).

repeat structure for humans (see Figure 13.9b). This nomenclature has now been adopted for all commercial STR kits typing DYS439.

NIST SRM 2395

The use of reference samples aids in obtaining calibrated and consistent results among laboratories performing Y-STR testing. The U.S. National Institute of Standards and Technology (NIST) released Standard Reference Material (SRM®) 2395 in July 2003 to serve as a human Y-chromosome DNA profiling standard. Five male DNA samples are provided with SRM 2395, each of which has been sequenced at over 20 different Y-STR loci including the common core loci present in all commercially available kits. An additional 21 loci are sequenced in a revised certificate released in October 2008. In the future, SRM 2391c will be certified at the 17 Y-STR loci present in Yfiler (as well as other loci should additional Y-STR kits become available).

Y-STR HAPLOTYPE DATABASES

A number of online Y-STR databases exist (Table 13.3). The forensic databases contain collections of anonymous individuals and can be used to estimate the frequency of specified

TABLE 13.3 Summary of Available Online Y-STR Databases (as of January 2011). For Updated Information, See http://www.cstl.nist.gov/biotech/strbase/y_strs.htm.

Database	Number of Samples (Population Groups)	Number of Y-STR Markers Tested	Website
Y-STR Haplotype Reference Database (YHRD)	91,601 (710 groups)	7 to 17	http://www.yhrd.org
US Y-STR Database (US Y-STR)	18,547 (5 groups)	11 to 17	http://www.usystrdatabase.org
Yfiler Haplotype Database	11,393 (13 groups)	17	http://www6.appliedbiosystems.com/yfilerdatabase/
Ysearch (for genealogists)	93,194	Up to 100	http://www.ysearch.org
Ybase (for genealogists)	16,059	Up to 49	http://www.ybase.org
Sorenson Molecular Genealogy (for genealogists)	37,588	7 to 37	http://www.smgf.org

Y-STR haplotypes. The genetic genealogy databases, such as Ysearch and Ybase, contain Y-STR haplotype information gathered by genetic genealogy companies with different sets of loci from males trying to make genealogical connections. Thus, the haplotypes in these genealogy databases are associated with specific individuals and family names.

YHRD

The largest and most widely used forensic and general population genetics Y-STR database, known as the Y-STR Haplotype Reference Database (YHRD), was created by Lutz Roewer and colleagues at Humbolt University in Berlin, Germany, and has been available online since 2000 (Roewer 2003, Willuweit & Roewer 2007). YHRD is internet-accessible at http://www.yhrd.org. As of January 2011, YHRD contains results from more than 89,000 samples with minimal haplotype loci results representing 710 different groups of sample submissions from various populations and countries around the world. Searches on YHRD may be conducted by population group or geographic location.

US Y-STR Database

A U.S. population-specific Y-STR Database (US Y-STR) was launched in December 2007 to enable haplotype frequency estimates on five different U.S. groups using the 11 SWGDAM recommended loci. The original version of US Y-STR contained 4796 African American profiles, 820 Asian, 5047 Caucasians, 2260 Hispanics, and 983 Native Americans. In some cases, further subdivision of these five primary groups can be examined if desired. An analysis of a version of the US Y-STR Database containing 7812 samples with full 17-locus Yfiler results found that 93.7% were distinct and 92.9% of these haplotypes were population-specific (Ge et al. 2010).

As of January 2011, US Y-STR had expanded to 18,547 SWGDAM profiles of which 8376 contained information at the 17 Yfiler loci. Where possible, US Y-STR has attempted to

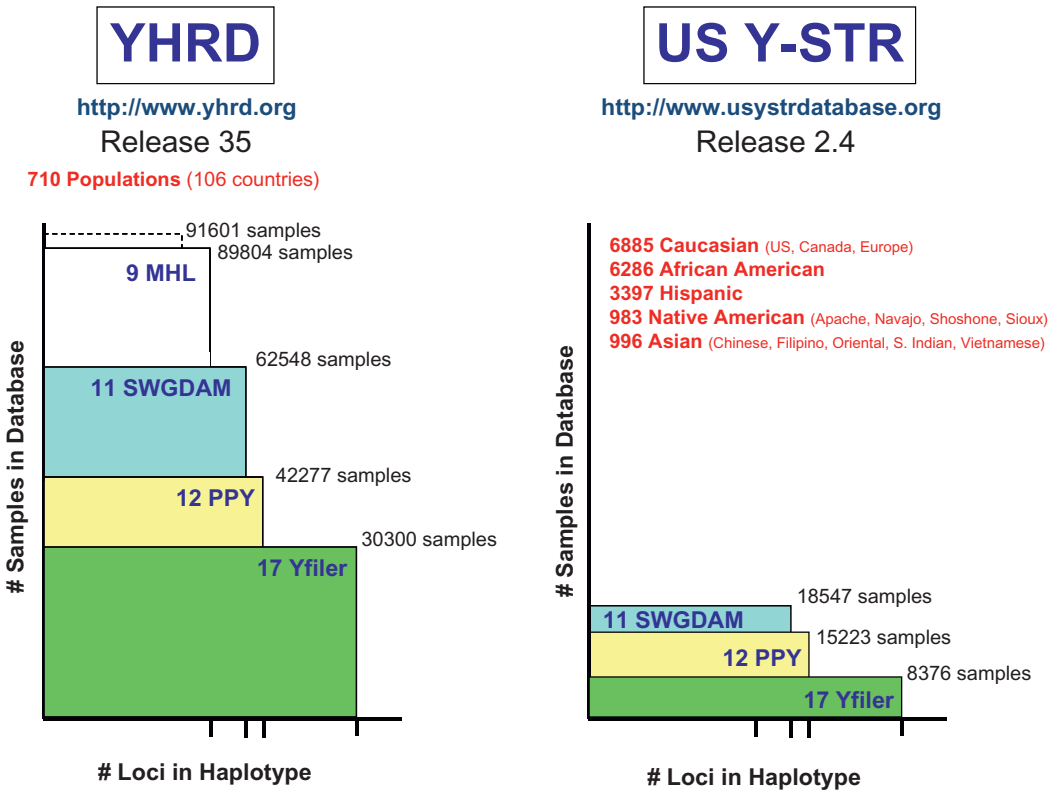


FIGURE 13.10 Comparison of YHRD and US YSTR database sample sizes and data content.

ensure that no duplicates are present through examining autosomal STR typing results on any samples possessing the same Y-STR profile. Having both autosomal and Y-STR data can be helpful in trying to sort out cases of common Y-STR haplotypes.

Figure 13.10 compares the numbers of samples containing various numbers of loci in YHRD and US Y-STR as of January 2011. When performing a search it is important to keep in mind that the denominator will change based on the loci selected in the search (see D.N.A. Box 13.1).

Genetic Genealogy Databases

Several genetic genealogy Y-STR haplotype databases are also available online (Table 13.3). These databases are typically not used for Y-STR forensic haplotype frequency estimates, but could be helpful in trying to associate a family surname with a particular haplotype if this information was desired in an investigation. These genetic genealogy databases contain information from the minimal haplotype loci, a subset of the minimal haplotype loci, or additional Y-STRs, and therefore cannot always be searched across all loci of interest.

D.N.A. BOX 13.1

A Y-STR PROFILE SEARCHED AGAINST SEVERAL HAPLOTYPE FREQUENCY DATABASES

The following profile was searched on January 15, 2011 against several databases (see Table 13.3): DYS19 (14), DYS389I (13), DYS398II (29), DYS390 (24), DYS391 (11), DYS392 (13), DYS393 (13), DYS385 a/b (11,15), DYS438 (12), DYS439 (13), DYS437 (15), DYS448 (19), DYS456 (17), DYS458 (18), DYS635 (23), and GATA-H4 (12).

The count and frequency of each of these searches is listed below.

The results shown here represent the counting method. Note that by adding additional loci the numerator goes down as the haplotype becomes more rare. Depending on the database and the type of haplotypes present (see Figure 13.10), the denominator can also become smaller as fewer profiles containing the requested information are being searched. As can be seen in the Yfiler column, there were no observations of the searched profile in the databases. The frequencies listed in this column are based on <1 observation in the size of the database.

The final column illustrates an upper bound 95% confidence interval for the frequency estimate when no observations are observed, which can be approximated by 3/N where N is the size of the database. The application of a 95% upper bound confidence interval on those results with at least one observation is covered in D.N.A. Box 13.2.

Database	Minimal Haplotype (9 loci)	SWGDAM (11 loci)	PowerPlex Y (12 loci)	Yfiler (17 loci)	3/N for Zero Observations
YHRD	403/89804 = 0.45%	29/62548 = 0.046%	14/42277 = 0.033%	0/30300 = <0.0033%	3/30300 = 0.0099%
US Y-STR	6/18547 = 0.032%	1/18547 = 0.0054%	1/15223 = 0.0066%	0/8376 = <0.012%	3/8376 = 0.036%
Yfiler database	64/11393 = 0.56%	4/11393 = 0.035%	4/11393 = 0.035%	0/11393 = <0.0088%	3/11393 = 0.026%

INTERPRETATION OF Y-STR RESULTS

Since the Y-chromosome is passed down unchanged (except for mutations) from father to son, the observation of a match with Y-STRs does not carry the same power of discrimination and weight in court as an autosomal STR match would. The lack of recombination between Y chromosome markers means that Y-STR results have to be combined into a haplotype for searching available databases as well as estimating the rarity of a particular haplotype.

Determining the Rarity of a Y-STR Profile

Generally speaking there are three possible interpretations resulting from comparing Y-STR haplotypes produced from question (Q) and known (K) samples: (1) *exclusion* because the Y-STR profiles are different and could not have originated from the same source, (2) *inconclusive* where there is insufficient data to render an interpretation or ambiguous results were obtained, or (3) *failure to exclude* (or *inclusion*) as the Y-STR haplotype results from the Q-K comparison are the same and could have originated from the same source.

When the Q and K samples (e.g., evidence and suspect haplotypes) do not match, then Y-STR typing is helpful in demonstrating the exclusion. However, estimating the strength of a match when a suspect's Y-STR haplotype cannot be excluded is more problematic because barring any mutations, paternal relatives (e.g., all brothers, male children, father, uncles, paternal grandfather, and paternal cousins) would be expected to have the same Y-STR profile (de Knijff 2003, see Figure 13.3).

Since it is common practice to place some significance on the likelihood of a random match with unrelated individuals, statistics derived from population data can be applied. Estimates for a random match with Y-STR haplotypes (and mtDNA sequence information) are done by the *counting method*, where the number of times the haplotype of interest is observed is divided by the total number of haplotypes in the database used. The size of the database used for the counting method makes a difference when trying to estimate the rarity of a Y-STR profile. The larger the number of unrelated individuals in the database (i.e., the denominator in the counting method calculation), the better the statistics will be for a random match frequency estimate.

D.N.A. Box 13.1 works through some example calculations with searches using subsets of the 17 Yfiler loci. This example explores the ability of different marker sets to resolve the Y-STR profile from unrelated lineages. When the full 17-locus profile was searched, it was not found in 30,300 17-locus haplotypes found in YHRD or in 8376 17-locus haplotypes located in US Y-STR.

An examination of 10,454 Yfiler profiles from 29 different population studies and the Yfiler haplotype database in 2006 found that approximately 95% of full 17-locus Yfiler profiles had never been observed before (Butler et al. 2007). In other words, these profiles were singletons in the population study or database. Frequency estimates calculated with the counting method, while not as powerful as those produced with unlinked autosomal STRs, may nevertheless be informative in many forensic casework scenarios and provide another piece of evidence in the overall framework of a case.

Charles Brenner, a forensic mathematician, has proposed the use of a probability estimate rather than a frequency estimate to judge the evidentiary value of a rare haplotype (Brenner 2010). Under the Brenner model, the fraction of singletons, or once-observed-haplotypes, in a data set become important to predict the rarity of future haplotypes that might be observed. For example, if 95% of 17-locus Y-STR profiles have never been observed before in a data set (see Butler et al. 2007), then a new 17-locus Y-STR profile being compared against this data set would likely not be observed 95% of the time. While this approach is not yet widely used, it has potential to help provide greater strength to rare haplotypes that are not present in current databases, such as the example shown in D.N.A. Box 13.1 with 17 loci.

SWGDAM Y-STR Interpretation Guidelines

The Scientific Working Group on DNA Analysis Methods (SWGDAM) approved Y-STR interpretation guidelines in July 2008 that were subsequently published in the January 2009 issue of *Forensic Science Communications* (SWGDAM 2009). These guidelines recommend that the loci on NRY should be considered linked as a single locus and that the source of the population database used in frequency estimation should be documented. Furthermore, searches of Y-STR haplotype frequency databases should be conducted using all loci for which results were obtained from the evidentiary sample. If there is less information on the known sample, then only those loci for which results were obtained from both the known and evidentiary sample should be used in the population database search.

SWGDAM endorsed the counting method with application of a confidence interval to correct for database size and sampling variation. The confidence interval equation commonly used for both Y-STR and mitochondrial DNA frequencies over the past decade assumes a normal distribution (Holland & Parsons 1999, Tully et al. 2000, SWGDAM 2009). At the January 2011 SWGDAM meeting, the Clopper-Pearson binomial model (Clopper & Pearson 1934) for calculating confidence intervals was discussed and advocated as being slightly more conservative than the normal distribution used previously (D.N.A. Box 13.2; see also D.N.A. Box 14.1).

Without providing a lot of guidance, the SWGDAM Y-STR Interpretation Guidelines also briefly address Y-STR mixtures, subpopulation (theta) corrections, and advocate joint match probabilities between Y-STRs and autosomal STR results (SWGDAM 2009).

Combining Y-STR Information and Autosomal DNA Results

In some cases, such as a fingernail scraping, a missing persons investigation, or a mass disaster reconstruction scenario, results from both Y-STR loci and a limited number of autosomal loci may be obtained. The question might then be asked can this information be combined to increase the rarity of a match since the autosomal data by itself may not be satisfactory? While this is a relatively new area and has not been investigated in detail yet, Sinha et al. (2004) reason that multiplication of the autosomal STR locus profile frequency obtained following the NRC II recommendations and the Y-STR haplotype frequency obtained with a minimal frequency threshold and correction for sampling (as demonstrated in the previous example) is still conservative based on lack of dependence between Y-STR loci and biological independence of chromosomes.

Bruce Walsh and colleagues at the University of Arizona have published an article expressing support for combining autosomal and Y-STR data (Walsh et al. 2008). This article was followed by a reminder that the approach should not be used with relatives (Amorim 2008).

Mutation Rates with Y-STR Markers

Several studies have examined mutation rates among the commonly used Y-STR loci. Most studies have focused on the minimal haplotype loci. Two different approaches

D.N.A. BOX 13.2

95% UPPER BOUND CONFIDENCE INTERVALS USING NORMAL AND CLOPPER-PEARSON APPROACHES

Confidence intervals may be used to reflect the reliability of a statistical estimate and are based on observed data. Provided that the statistical model is correct, confidence intervals are intended to offer assurance that all data obtained with a procedure, or that might be obtained in the future, should include the true value of the parameter the proportion of time set by the confidence level. A 95% confidence interval is most commonly calculated.

When a Y-STR haplotype (or mtDNA sequence) has not been observed in a database of size N, the 95% confidence interval is

$$1 - (0.05)^{1/N}$$

This value is very close to 3/N, which was used in the D.N.A. Box 13.1 example calculations.

An upper bound 95% confidence interval can be placed on a profile's frequency using

$$p + 1.96\sqrt{\frac{(p)(1-p)}{N}}$$

where the frequency (p) is determined from the number of observations (x) in a database containing N profiles. This "normal" approximation interval is the simplest formula to calculate and has been widely used (see Holland & Parsons 1999, Tully et al. 2001) but is known to be problematic in situations with small sample sizes or very few observations.

The Clopper-Pearson formula, named after the authors of the paper describing it in 1934, provides a more conservative value for confidence intervals when very low counts are observed from a haplotype database. The formula for the upper 95 percent confidence limit using Clopper-Pearson is

$$\sum_{k=0}^{x}\binom{N}{k}p_0^k(1-p_0)^{N-k} = 0.05$$

where N = the database size, x = the number of observations of the haplotype in the database, $k = 0, 1, 2, 3 \ldots x$ observations, and p = the haplotype frequency at which x or fewer observations are expected to occur 5% of the time. This cumulative binomial distribution formula is solved for p through serial iterations and therefore requires the use of a computer program. In the examples below, an Excel spreadsheet from Steven

Count Values (see D.N.A. Box 13.1)	Frequency p = x/N	Normal 95% Confidence Interval	Clopper-Pearson 95% Confidence Interval
YHRD 9 loci: **403**/89804	0.449%	0.485%	0.487%
YHRD 12 loci: **14**/42277	0.0331%	0.0477%	0.0518%
US Y-STR 12 loci: **1**/15223	0.0657%	0.0174%	0.0317%

D.N.A. BOX 13.2 *(cont'd)*

Myers (California Department of Justice) was utilized.

Note that with a large number of observations, such as 403 out of a database of 89,804, there is almost no difference between the normal and Clopper-Pearson approaches. However, the normal method is less conservative (i.e., provides a more rare frequency) when the haplotype frequency is low, such as 1 out of 15,223 or even 14 out of 42,277. Although there are differences in these calculations, re-evaluation by the Clopper-Pearson method will not suddenly change a reported result by orders of magnitude or likely change the outcome of a report significantly.

In March 2010 the US Y-STR database changed its 95% confidence interval calculations to the Clopper-Pearson method.

Sources:

Clopper, C.J., & Pearson, E.S. (1934). The use of confidence or fiducial limits illustrated in the case of the binomial. Biometrika, *26, 404–413.*

HaploCALc_1.0 Excel spreadsheet kindly provided by Steven P. Myers, California Department of Justice.

Holland, M.M., & Parsons, T.J. (1999). Mitochondrial DNA sequence analysis – validation and use for forensic casework. Forensic Science Review, *11, 21–50.*

Tully, G., et al. (2001). Considerations by the European DNA profiling (EDNAP) group on the working practices, nomenclature and interpretation of mitochondrial DNA profiles. Forensic Science International, *124, 83–91.*

http://en.wikipedia.org/wiki/Binomial_proportion_confidence_interval and http://en.wikipedia.org/wiki/Confidence_interval.

have been used: deep-rooted pedigrees (Heyer et al. 1997, Bonne-Tamir et al. 2003) and male germ-line transmissions from confirmed father/son pairs (Bianchi et al. 1998, Kayser et al. 2000, Dupuy et al. 2001, Dupuy et al. 2004, Kurihara et al. 2004, Gusmão et al. 2005, Budowle et al. 2005, Decker et al. 2008, Ge et al. 2009).

The pedigree approach has the advantage of not having to run as many samples but when differences are seen it can be hard to attribute the mutation to the proper generation (see Bonne-Tamir et al. 2003) or to potential illegitimacy (Heyer et al. 1997). Heyer et al. (1997) tested only 42 males but were able to infer information from 213 generations or meioses (once three illegitimate lines had been removed) while Bonne-Tamir et al. (2003) examined 74 male samples that spanned 139 generations. Of course the pedigree approach requires detailed genealogical records and no breakdown in the paternal lineages through illegitimacy.

The mutation rates for Y-STRs are in the same range as autosomal STRs, namely around one to four per thousand generational events (0.1% to 0.4%) (see Table 13.2). Only DYS458 and DYS439 mutate faster than 0.4% based on current data. A compilation of the various studies reveals that compound repeat locus DYS458 is the most likely to mutate with DYS438 being the least likely to change. As with autosomal STRs, single-repeat changes are favored over multiple-repeat jumps. Allele gains are more common than allele losses as the mutations occur with not only locus-specific but also allele-specific differences in mutation rate (Dupuy et al. 2004). Mutations typically only occur when 11 or more homogeneous repeats are immediately adjacent to one another (Kayser et al. 2000).

Kayser and Sajantila (2001) discuss the implications of mutations for paternity testing and forensic analysis. They observed mutations at two Y-STRs within the same father/son pair suggesting that differences at three or more Y-STRs are needed before an "exclusion" can be declared with paternity testing or kinship analysis, which is typically the same criteria used for paternity testing with autosomal loci. However, a recent study found a single instance of three mutations out of 17 Y-STRs tested in a confirmed father-son pair (Goedbloed et al. 2009).

Occasionally duplications or even triplications of a Y-STR locus have been reported, particularly for DYS19 (Butler et al. 2005). It is important to keep this fact in mind so that two peaks at the DYS19 locus are not automatically interpreted as coming from a mixture of two males. Both of these issues, namely mutations impacting paternity analysis and duplications of loci potentially confusing mixture interpretation, suggest that analysis of additional Y-STR loci can be helpful in these situations.

Y-STR USE IN FORENSIC CASEWORK

Y-STR assays have been used for several years on a limited basis to aid forensic casework. Their use has been much more widespread in Europe than the United States although a greater number of U.S. labs are now embracing Y-STR testing. Early work in the U.S. with Y-STRs was performed in the late 1990s by the New York City Office of the Chief Medical Examiner (OCME). ReliaGene Technologies, Inc. (New Orleans, LA) developed the first Y-STR kit, Y-PLEX 6, and started doing Y-chromosome testing in late 2000.

The New York City Office of the Chief Medical Examiner primarily uses Y-STR testing when any one of the four scenarios is met (Prinz 2003): (1) evidence is positive for semen but no DNA foreign to the victim can be detected, or potential male alleles are below the detection threshold with autosomal STR tests; (2) the evidence in question is amylase positive and a male/female mixture is expected; (3) a large number of semen stains need to be screened; and (4) the number of semen donors needs to be determined (e.g., suspected gang rape).

Determining the amount of male DNA present rather than the total amount of male and female DNA is important for getting on-scale results with Y-STR testing. One approach is to estimate the general level of male DNA present by assessing the strength of the p30 antibody signal (see Chapter 1) (Prinz 2003). The recent availability of real-time PCR assays specific to the male DNA component of a forensic mixture (e.g., Quantifiler Y kit) provides a more high-tech approach.

There have been several published reports describing the use and value of Y-STR testing in forensic casework. Some of these published results are summarized in Table 13.4. ReliaGene reported use of Y-STRs on 188 forensic samples from 2000 to 2003 with their Y-PLEX 6 and Y-PLEX 5 kits (Sinha et al. 2004). Samples were from epithelial cells including azospermic seminal fluid, sweat or saliva, sperm, fingernails, blood, and other tissues. Y-STR testing has been accepted in several jurisdictions throughout the United States (Sinha et al. 2004). Since Y-STR kits became more available in 2003 (PowerPlex Y) and 2004 (Yfiler), this information has been used in a growing number of cases to aid forensic investigations.

TABLE 13.4 Some Published Reports Describing Use of Y-STRs in Forensic Casework.

Kit/Loci Used	Reference	Comments
In-house assay with DYS19, DYS390, DYS389I/II	Prinz et al. (2001)	In one year at the New York City Office of the Chief Medical Examiner, Y-STR testing was performed in more than 500 cases with over 1000 evidence and reference samples examined. A full or partial profile was obtained on 81% of all tested evidence samples (740 worked/915 samples tested). Mixtures of at least two males were observed in 97 instances. In male/female mixtures of up to 1:4000, the male component could be cleanly detected.
In-house assay with 9 Y-STR loci amplified in 3 PCR reactions	Dekairelle & Hoste (2001)	Y-STR typing was attempted on 166 semen traces from 89 cases that failed to yield a detectable male autosomal profile following differential extraction. About half of the cases had sufficient DNA to produce a Y-STR profile.
In-house assay with DYS393, DYS389I/II	Sibille et al. (2002)	Y-STR results could still be obtained more than 48 hours after the sexual assault in 30% of the cases examined. In 104 swabs collected with no evidence of sperm, Y-STRs could be detected in ~29% of the samples tested.
In-house assay with DYS19, DYS390, DYS389I/II	Prinz (2003)	Six case studies are reviewed along with advantages and disadvantages of Y-STR testing in each case: (1) different semen donors on vaginal swab and underwear; (2) possible oligospermic perpetrator gave a nice Y-STR profile but failed to have a "male" fraction with differential extraction; (3) oral intercourse with no autosomal results—not possible to enrich male cell fraction with differential extraction in cases involving saliva; (4) presence of multiple semen donors created a complex autosomal mixture that could be sorted out with Y-STR results; (5) sperm cell fraction lacked amelogenin Y-specific peak due to known deletion—Y-STR results confirmed that the sperm cell fraction DNA was of male origin; and (6) Y-STR testing was used to rapidly screen 18 semen stains for comparison to 5 suspects and thus save the time of performing the differential extraction.
Y-PLEX 6 and Y-PLEX 5 kits	Sinha (2003)	Five cases are reviewed: (1) criminal paternity case with a male fetus where the alleged father could not be excluded as the biological father; (2) autosomal STR test resulted in an uninterpretable mixture—suspect was excluded at 3 of the 7 Y-STR loci tested; (3) Y-PLEX 6 STR profile matched suspect with sweat stains on cloth found at crime scene; (4) fingernail cuttings from a victim matched a suspect at 11 Y-STR loci while another suspect was excluded at 2 loci; (5) semen positive stain with no sperm cells produced a Y-PLEX 6 profile consistent with the male suspect.
Y-PLEX 6 and Y-PLEX 5 kits	Sinha et al. (2004)	Seven cases are reviewed (some are the same as Sinha 2003) and a list of cases where Y-STR results have been accepted in U.S. courts is provided.

ChrY Locus Duplications or Triplications

Due to the palindromic sequences often found on ChrY, Y-STR loci can sometimes exhibit as duplications or even triplications. These multi-allelic patterns may be inherited or passed on across generations as has been observed when studying father-son sample pairs (Decker et al. 2008). ChrY duplications, which occur in single-source samples, can confuse or complicate mixture interpretation if scientists are not aware of this phenomenon (Butler et al. 2005). Sequence regions flanking STR regions may also be duplicated elsewhere in the Y-chromosome and may create artifacts when PCR primers are moved during multiplex assay design (Butler & Schoske 2004).

Duplications or triplications of several Y-STRs have been reported for DYS19, DYS390, and DYS391. For example, one study found nine duplications for DYS19 in 7772 individuals (Kayser et al. 2000). Triplicated DYS385 alleles have also been reported (Kayser et al. 2000, Butler et al. 2002, Kurihara et al. 2004). These possible multi-allelic patterns need to be kept in mind so that a mixture is not expected when encountering multiple alleles at a single locus that could legitimately come from a single-source sample.

ChrY Deletions

Just as ChrY duplications may occur, deletions also exist in some individuals. An examination of father and son sample pairs found that these deletions may be inherited (Decker et al. 2008). Sometimes the deletions from the Y-chromsome can be greater than one megabase in size (Takayama et al. 2009). Deletions of the amelogenin Y region (see Chapter 5) can result in DYS458, the closest Y-STR locus (see Figure 13.6), being missing from a Yfiler profile.

Studies with Additional Y-STR Markers

Within the last several years, a number of new Y chromosome STR markers have been characterized and new multiplex assays developed (Prinz et al. 1997, Butler et al. 2002, Redd et al. 2002, Schoske et al. 2003, Hanson & Ballantyne 2007). Information on additional Y-STR loci and assays is available on the NIST STRBase website at http://www.cstl.nist.gov/biotech/strbase/y_str.htm.

A few population studies have been conducted that go beyond the minimal haplotype loci in order to assess the power of additional markers in resolving most common types. For example, Berger et al. (2003) found that addition of the multi-copy marker DYS464 to the minimal haplotype loci increased the number of different haplotypes in a set of 135 Austrian males from 110 to 122. Schoske and co-workers (2004) demonstrated that 25 samples, which possessed an indistinguishable most common minimal haplotype could be subdivided into 24 different groups (only one pair could not be resolved) with the addition of DYS438, DYS439, DYS464, DYS458, DYS460, and DYS437. Studies have been performed to compare various combinations of Y-STR loci to the minimal haplotype in order to determine the best order in which to apply the markers (Alves et al. 2003). Thus, other loci beyond the core minimal haplotype or SWGDAM-recommended loci are likely to play a valuable role with future forensic DNA analysis involving the Y-chromosome.

Table 13.5 demonstrates that additional loci can be helpful in subdividing shared common haplotypes. The minimal haplotype 9 loci, the SWGDAM 11 loci, the PowerPlex Y 12

TABLE 13.5 Numbers of Unique (Blue Font) and Shared Haplotypes Observed with Various Combinations of YSTR Loci across 656 U.S. Population Samples That Are Part of the Yfiler Haplotype Database (Butler et al. 2007). The Shared Haplotypes with Yfiler and ALL 37 are Highlighted in Red Font. Data Used for This Analysis Is Available at http://www.cstl.nist.gov/biotech/strbase/NISTpop.htm.

# Times Haplotype Observed	MHL	SWGDAM	PPY	Yfiler	ALL 37
1	429	486	505	626	652
2	34	33	34	12	2
3	13	10	14	2	.
4	4	6	3	.	.
5	3	1	2	.	.
6	1	1	.	.	.
7	1	2	1	.	.
8	1	.	.	.	.
9	2	.	.	.	.
10	.	1	.	.	.
11	1	.	.	.	.
12	.	.	1	.	.
13	1	.	.	.	.
14	.	.	.	.	.
15	.	1	.	.	.
16	.	.	.	.	.
17	.	.	.	.	.
18	.	.	.	.	.
19	.	.	.	.	.
20	.	.	.	.	.
21	.	.	.	.	.
22	.	.	.	.	.
23	.	.	.	.	.
24	.	.	.	.	.
25	.	.	.	.	.
26	1	.	.	.	.
Haplotype Diversity	0.997	0.998	0.999	0.999	0.999
Discrimination Capacity	0.748	0.825	0.854	0.976	0.997
Total Number of Haplotypes	491	541	560	640	654

loci, and the Yfiler 17 loci as well as an additional 20 Y-STRs (for a total of 37) (Butler et al. 2006, Decker et al. 2007) were examined to look for the number of unique and shared haplotypes (Table 13.5). The first column shows that 26 of the 656 samples examined possess a "most common type" 9-locus haplotype that is subdivided with additional loci. It is evident that the number of unique haplotypes increases as additional loci are used and that the number of samples sharing haplotypes is also reduced.

With the PowerPlex Y 12 loci, there are 505 unique haplotypes and the most common type is only shared in 12 individuals. The five additional loci present in Yfiler give rise to 626 unique types, 12 haplotypes that are shared twice, and two haplotypes that were observed three times (Table 13.5). At the 17-locus Yfiler level, 95% of this data set are resolved from one another (i.e., are singletons). Using Yfiler, the most common type that was shared in 12 individuals with the 12 PowerPlex Y loci is subdivided into 9 unique types and one shared by the remaining three individuals.

In the data set shown in Table 13.5, there is some information gained by going beyond the Yfiler loci but still two sample pairs could not be subdivided. These individuals could very well be paternal relatives (the samples are anonymous). These results also illustrate that there can be diminishing returns when examining additional loci.

Looking across multiple studies, the best additional Y-STRs for resolving common 9-locus, 12-locus, or 17-locus haplotypes appear to include DYS449, DYS481, DYS570, and DYS576 (Decker et al. 2007; Hanson & Ballantyne 2007; Rodig et al. 2007). One recent study utilized 14 additional Y-STRs in a single multiplex to subdivide all 8 remaining shared haplotypes found following analysis of 17-locus Yfiler types from 572 U.S. Caucasians and African Americans (Hanson & Ballantyne 2007). The 14 Y-STRs used were DYS444, DYS446, DYS449, DYS459a/b, DYS481, DYS508, DYS522, DYS527a/b, DYS549, DYS552, DYS570, DYS576, DYS607, and DYS627.

A group of scientists lead by Manfred Kayser from Erasmus University in The Netherlands examined 186 Y-STR markers in nearly 2000 DNA-confirmed father-son pairs (Ballantyne et al. 2010). They found that by using a set of 13 of the most mutable Y-STRs it was possible to distinguish distantly and even closely related males. The potential may exist in the future to individualize males rather than just male lineages using a battery of Y-STRs possessing high mutation rates.

Y-SNPS AND HAPLOGROUPS

Single nucleotide polymorphisms, *Alu* insertions, and insertion/deletion markers exist on the Y-chromosome just as they do throughout the rest of the human genome (see Chapter 12). Most of the focus to date in forensic DNA typing applications has been on Y-STRs rather than Y-SNPs due to the higher power of discrimination with the multi-allelic Y-STRs. However, Y-SNPs play an important role in human migration studies because they enable effective evaluation of major differences between population groups.

Y-SNP alleles are typically designated as either "ancestral" or "derived" and can be recorded in a simple binary format of 0 or 1 for ancestral and derived, respectively. The ancestral state of a Y-SNP marker is usually determined by comparison to a chimpanzee DNA sequence for the same marker (Underhill et al. 2000).

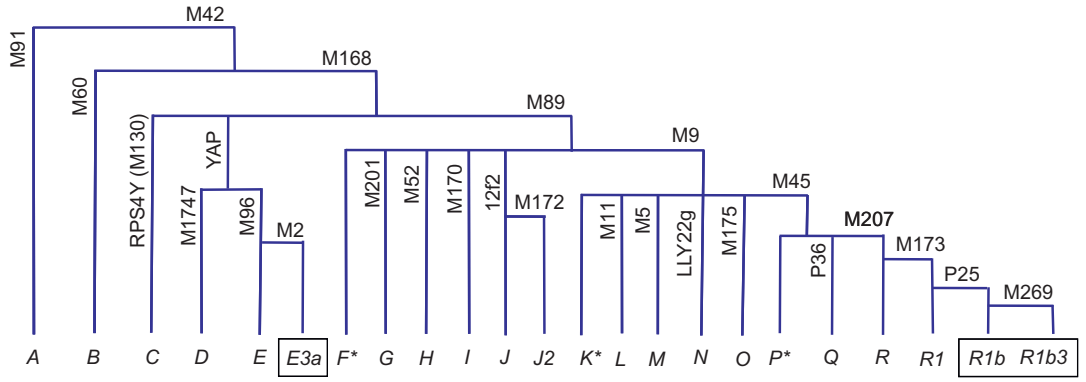


FIGURE 13.11 The original 2002 Y Chromosome Consortium Tree with 18 major haplogroups (A-R). Representative Y-SNP markers that define each haplogroup are listed next to the branch point. The most common African American haplogroup is E3a. The most common Caucasian (European) haplogroup is R1b/R1b3. Examination of additional samples and markers since 2002 has expanded the branches on this tree. For an up-to-date version, see http://www.isogg.org/tree/.

The Y Chromosome Consortium Haplogroup Tree

The Rosetta Stone for interpreting the plethora of Y-chromosome haplogroups listed in the literature was published by the Y Chromosome Consortium (YCC) in the February 2002 issue of *Genome Research* (YCC 2002) and updated in 2008 (Karafet et al. 2008). The YCC is an international group of scientists lead by Michael Hammer from the University of Arizona, Peter Underhill from Stanford University, Mark Jobling from Leicester University, and Chris Tyler-Smith who at that time was at Oxford University. Their paper entitled "A nomenclature system for the tree of human Y chromosomal binary haplogroups" opened the way for an easier understanding of seven previously published methods for describing information from the same SNP markers. The "YCC tree" as it is commonly called describes the position of almost 250 bi-allelic markers in differentiating 153 different haplogroups (YCC 2002).

A slightly modified and updated YCC tree was published in August 2003 (Jobling and Tyler-Smith 2003) and is regularly updated by a group of avid genetic genealogists who are part of the International Society of Genetic Genealogy (ISOGG). The YCC published an updated tree in the May 2008 issue of Genome Research (Karafet et al. 2008). With more than 600 Y-SNPs available on worldwide population data sets, the YCC tree expanded.

Figure 13.11 highlights the major branches of the 2002 YCC tree along with some of the Y-SNP markers that help define the various branches. For example, observation of the derived allele for M2 in a sample classifies it into the 2002 E3a haplogroup. Y-chromosome haplogroup designation and characterization has greatly benefited from the YCC tree.

Before 2002, if a "G" (derived state) was observed in a sample when typing the M2 (sY81 or DYS271) marker, then the sample could be reported as belonging to haplogroup (Hg) 8 by Jobling's nomenclature (Jobling & Tyler-Smith 2000), Hg III by Underhill's naming procedure (Underhill et al. 2000), or Hg 5 by Hammer's description (Hammer et al. 2001). On the YCC tree, M2 derived alleles define the Hg E3a. Needless to say, the unified and universal nomenclature is much easier to understand and permits comparisons of results across laboratories.

A number of the SNP typing technologies reviewed in Table 12.3 have also been used for Y-SNP typing (Butler 2003). Two of the more popular SNP typing methodologies have been allele-specific primer extension (ASPE) and allele-specific hybridization (ASH). Vallone and Butler (2004) observed complete concordance comparing ASPE and ASH in almost 4000 Y-SNP allele calls.

Y-SNPs have been useful in demonstrating some admixture in U.S. populations (Hammer et al. 2006) and have helped confirm self-declared ethnicities (Lao et al. 2010). Next-generation sequencing technology has been used to identify Y-SNPs between ancestral lineages (Xue & Tyler-Smith 2010).

ChrY HISTORICAL AND GENEALOGICAL STUDIES

Genealogists in large numbers are beginning to turn to Y-chromosome DNA testing to extend their research efforts (Brown 2002). Hundreds of thousands of genetic genealogy tests have been performed in the past few years. Family Tree DNA (Houston, TX) is probably the most progressive genetic genealogy company and offers routine tests for 67 Y-STRs with options to obtain further information from ChrY if desired.

Y-chromosome testing has played a role in addressing some interesting historical questions, such as the Thomas Jefferson-Sally Hemings affair (Foster et al. 1998), the genetic legacy of the Mongols (Zerjal et al. 2003), and the Romanov children identification (Coble et al. 2009). In addition, dozens of Y-chromosome markers are used to aid efforts in genealogical family history research. DNA information is only part of the evidence available in most investigations and should be considered carefully in the context of the "case" without overstepping the bounds of conclusions that can be drawn.

Genetic genealogy using Y-chromosome STR markers in conjunction with surname studies originated with a study published by Bryan Sykes in 2000 (Sykes & Irven 2000). Using four Y-STRs, DYS19, DYS390, DYS391, and DYS393, Sykes tested 48 men bearing the surname "Sykes" sampled from several regions of England. Of the 48 tested, 21 of them exhibited the core Sykes haplotype and several others were only one mutational step away from the core haplotype. Sykes interpreted these results to reflect a common origin coming from an ancestor that lived about 700 years ago. While some interesting connections are being made with DNA to aid genealogical research, results should be interpreted with caution.

SUMMARY

The field of Y-chromosome analysis and its application to human identity testing has undergone rapid improvements in recent years. New markers and population groups have been characterized. Commercial kits are now available and beginning to be used in greater numbers to aid forensic cases. Y-STR haplogroup databases are growing in size enabling greater statistical power with the counting method. In addition to benefiting forensic casework, ChrY testing is also aiding familial searching efforts in helping to screen out adventitious matches due to autosomal allele sharing (see Appendix 2). The future may involve more interaction between the application of genetic genealogy and forensic science to help bring cases to closure without prior suspects.

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